

(see Figure 25.16). Roughly 200 million years ago, Pangaea began to break apart; by 20 million years ago, the continents were within a few hundred kilometers of their present locations.

We can use our understanding of evolution and continental drift to predict where fossils of different groups of organisms might be found. For example, scientists have constructed evolutionary trees for horses based on anatomical data. These trees and the ages of fossils of horse ancestors suggest that the genus that includes present-day horses (*Equus*) originated 5 million years ago in North America. Geologic evidence indicates that at that time, North and South America were not yet connected, making it difficult for horses to travel between them. Thus, we would predict that the oldest *Equus* fossils should be found only on the continent on which the group originated—North America. This prediction and others like it for different groups of organisms have been upheld, providing more evidence for evolution.

Evolution also explains biogeographic data. For example, a group of three species of freshwater fish in the family Galaxiidae are found in regions separated by wide stretches of open ocean (Figure 22.22). None of these three species can

survive in salt water, so how can we explain their geographic distributions? These three species represent all known living descendants of a common ancestor. Since all of the living species cannot survive in salt water, it is likely that their common ancestor also was restricted to freshwater habitats. At first glance, this only deepens the mystery—how could that ancestor or its descendants have swum across vast stretches of ocean? A clue comes from genetic analyses indicating that the evolutionary lineages leading to these three species began to diverge about 55 million years ago. By that time, the southernmost portion of Pangaea was breaking apart into what would eventually become South America, Australia, and Antarctica. This suggests that their ancestors had spread across southern Pangaea before it broke apart. At later times, these three freshwater species originated on the continental regions where they currently live, resulting in their present-day geographic distribution.

What Is Theoretical About Darwin's View of Life?

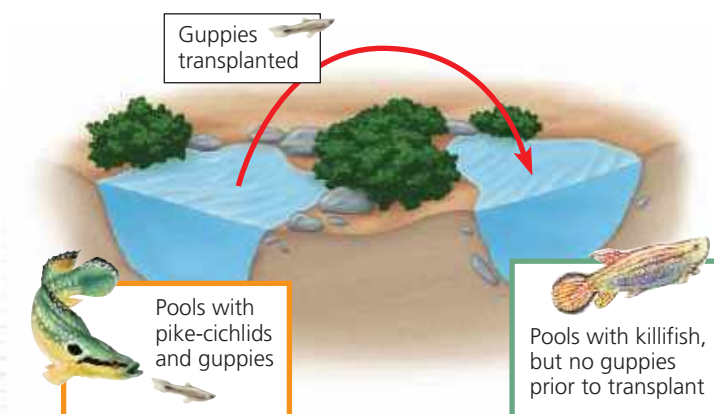
Some people dismiss Darwin's ideas as “just a theory.” But the *pattern* of evolution—the observation that life has evolved

Scientific Skills Exercise

Making and Testing Predictions

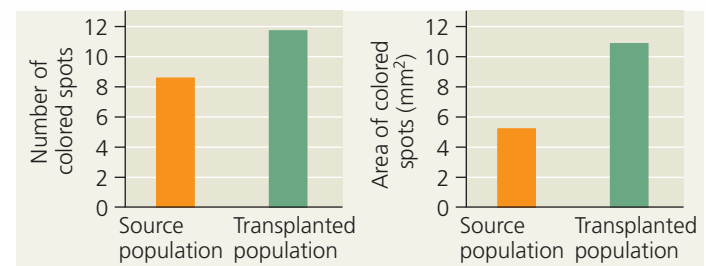
Can Predation Result in Natural Selection for Color Patterns in Guppies? Our understanding of evolution changes constantly as new observations lead to new hypotheses—and hence to new ways to test our understanding of evolutionary theory. Consider the wild guppies (*Poecilia reticulata*) that live in pools connected by streams on the Caribbean island of Trinidad. Male guppies have highly varied color patterns that are controlled by genes that are only expressed in adult males. Female guppies choose males with bright color patterns as mates more often than they choose males with drab coloring. But the bright colors that attract females also can make the males more conspicuous to predators. Researchers observed that in pools with few predator species, the benefits of bright colors appear to “win out,” and males are more brightly colored than in pools where predation is more intense.

One guppy predator, the killifish, preys on juvenile guppies that have not yet displayed their adult coloration. Researchers predicted that if adult guppies with drab colors were transferred to a pool with only killifish, eventually the descendants of these guppies would be more brightly colored (because of the female preference for brightly colored males).



How the Experiment Was Done Researchers transplanted 200 adult guppies from pools containing pike-cichlid fish, predators of adult guppies, to pools with only killifish, predators of juvenile guppies. They tracked the number of bright-colored spots and the total area of those spots on adult male guppies in each generation.

Data from the Experiment After 22 months (15 generations), researchers compared the color pattern data for guppies from the source and transplanted populations.



Data from J. A. Endler, Natural selection on color patterns in *Poecilia reticulata*, *Evolution* 34:76–91 (1980).

INTERPRET THE DATA

- Identify the following elements of hypothesis-based science in this example: **(a)** question, **(b)** hypothesis, **(c)** prediction, **(d)** control group, and **(e)** experimental group. (For additional information about hypothesis-based science, see Concept 1.3 and the Scientific Skills Review in Appendix D.)
- Explain how the types of data the researchers chose to collect enabled them to test their prediction.
- What conclusion do you draw from the data presented above?
- (a)** Predict what would happen if, after 22 months, guppies from the transplanted population were returned to the source pool. **(b)** Describe an experiment to test your prediction.

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.